Digital Electronic Circuits (EC2L004)

Lecture 3: Binary Numbers – Part III

Learning Objectives:

- Signed numbers
- Binary Codes
- Binary Logic
- Binary Arithmetic
- Miscellaneous

Signed Binary Numbers

Arithmetic addition

- The addition of two numbers in the signed-magnitude system follows the rules of ordinary arithmetic.

- **If the signs are the same**, we add the two magnitudes and give the sum the common sign.

- **If the signs are different**, we subtract the smaller magnitude from the larger and give the difference the sign if the larger magnitude.

Signed Binary Numbers

Arithmetic addition

- The addition of two signed binary numbers with negative numbers represented in signed-2's-complement form is obtained from the addition of the two numbers, including their sign bits.

- A carry out of the sign-bit position is discarded.

Signed Binary Numbers

- Arithmetic addition

+ 6	00000110	− 6	11111010
+13	00001101	+13	00001101
+ 19	00010011	+ 7	00000111
+ 6	00000110	− 6	11111010
−13	11110011	−13	11110011
− 7	11111001	− 19	11101101

Signed Binary Numbers

- Arithmetic Subtraction

1. Take the 2's complement of the subtrahend (including the sign bit) and add it to the minuend (including sign bit).

2. A carry out of sign-bit position is discarded.

$$(\pm A) - (+B) = (\pm A) + (-B)$$
$$(\pm A) - (-B) = (\pm A) + (+B)$$

Example:

(-6) –(-13) → (11111010 – 11110011)

→ (11111010 + 00001101)

→ 00000111 (+ 7)

Signed Binary Numbers

- It is worth noting that binary numbers in the signed-complement system are added and subtracted by the same basic addition and subtraction rules as unsigned numbers.

- Therefore, **computers need only one common hardware circuit to handle both types of arithmetic.**

Binary Codes

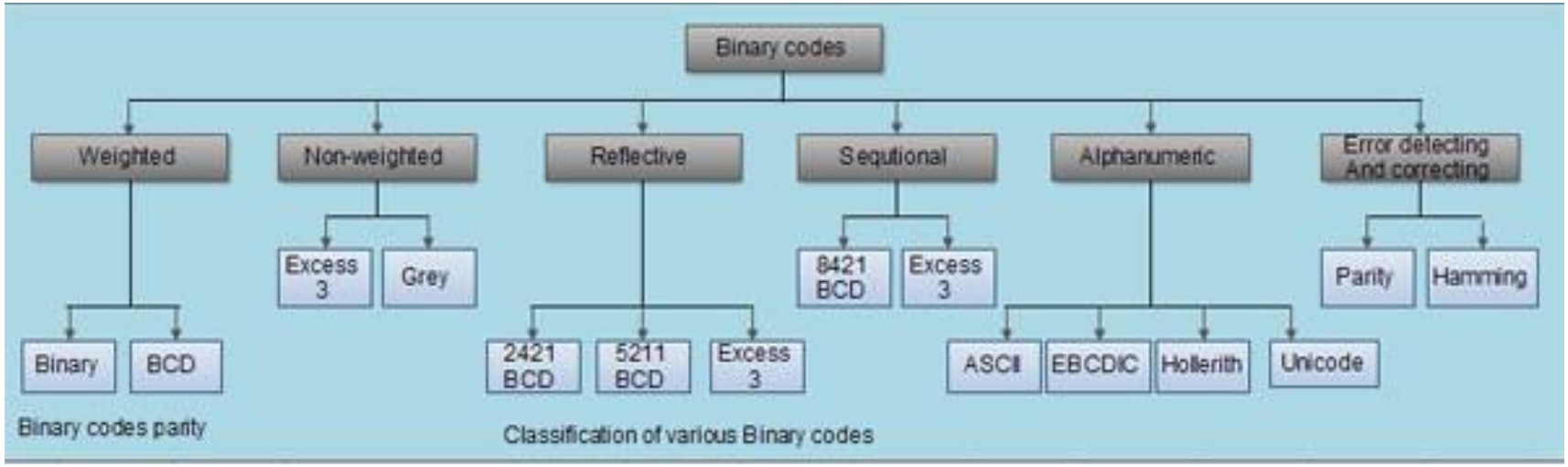
Binary codes

- Digital data is represented, stored and transmitted as groups of binary digits also known as binary code.

- An n-bit binary code is a group of "n" bits that assumes up to 2^n distinct combination of 1's and 0's, with each combination representing one element of the set that is being coded.

- Example Binary codes:
  - Binary coded decimal (BCD)
  - 2421
  - Excess-3
  - 84-2-1 code
  - Gray code

Binary Codes



- **Weighted codes:** In weighted codes, each digit is assigned a specific weight according to its position.

- **Non-weighted codes:** non-weighted codes are not appositionally weighted.

Binary Coded Decimal Code

- A number with k decimal digits will require $4k$ bits in BCD.
- Decimal 396 is represented in BCD with 12bits as 0011 1001 0110, with each group of 4 bits representing one decimal digit.
- A decimal number in BCD is the same as its equivalent binary number only when the number is between 0 and 9.
- The binary combinations 1010 through 1111 are not used and have no meaning in BCD.

Binary-Coded Decimal (BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Binary Coded Decimal Code

- Example: Consider decimal 185 and its corresponding value in BCD and binary:

$$(185)_{10} = (0001\ 1000\ 0101)_{BCD} = (10111001)_2$$

Other Decimal Codes

Four Different Binary Codes for the Decimal Digits

Decimal Digit	BCD 8421	2421	Excess-3	8, 4, −2, −1
0	0000	0000	0011	0000
1	0001	0001	0100	0111
2	0010	0010	0101	0110
3	0011	0011	0110	0101
4	0100	0100	0111	0100
5	0101	1011	1000	1011
6	0110	1100	1001	1010
7	0111	1101	1010	1001
8	1000	1110	1011	1000
9	1001	1111	1100	1111

Gray Code

- The advantage is that only bit in the code group changes in going from one number to the next.

Gray Code	Decimal Equivalent
0000	0
0001	1
0011	2
0010	3
0110	4
0111	5
0101	6
0100	7
1100	8
1101	9
1111	10
1110	11
1010	12
1011	13
1001	14
1000	15

ASCII Character Code

- American Standard Code for Information Interchange (ASCII)

American Standard Code for Information Interchange (ASCII)

	$b_7b_6b_5$							
$b_4b_3b_2b_1$	000	001	010	011	100	101	110	111
0000	NUL	DLE	SP	0	@	P	`	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	"	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	&	6	F	V	f	v
0111	BEL	ETB	'	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	,	<	L	\	l	|
1101	CR	GS	–	=	M	]	m	}
1110	SO	RS	.	>	N	^	n	~
1111	SI	US	/	?	O	–	o	DEL

ASCII Character Codes and Properties

- ASCII has some has interesting properties:
  - Digits 0 to 9 span Hexadecimal values 30_{16} to 39_{16}
  - Upper case A-Z span 41_{16} to $5A_{16}$
  - Lower case a-z span 61_{16} to $7A_{16}$
    - Lower to upper case translation (and vice versa) occurs by flipping bit 6.

Binary Logic

Binary Logic

- Binary logic consists of binary variables and a set of logical operations.

- The variables are designated by letters of the alphabet, such as *A, B, C, x, y, z,* etc., with each variable having two and only two distinct possible values: 1 and 0.

- There are three basic logical operations: AND, OR, and NOT. Each operation produces a binary result, denoted by *z.*

Binary logic (cont'd…)

1. AND: This operation is represented by a dot or by the absence of an operator. For example, $x \cdot y = z$ or $xy = z$ is read "x AND y is equal to z." The logical operation AND is interpreted to mean that $z = 1$ if and only if $x = 1$ and $y = 1$; otherwise $z = 0$. (Remember that x, y, and z are binary variables and can be equal either to 1 or 0, and nothing else.) The result of the operation $x \cdot y$ is z.

2. OR: This operation is represented by a plus sign. For example, $x + y = z$ is read "x OR y is equal to z," meaning that $z = 1$ if $x = 1$ or if $y = 1$ or if both $x = 1$ and $y = 1$. If both $x = 0$ and $y = 0$, then $z = 0$.

3. NOT: This operation is represented by a prime (sometimes by an overbar). For example, $x' = z$ (or $\overline{x} = z$) is read "not x is equal to z," meaning that z is what x is not. In other words, if $x = 1$, then $z = 0$, but if $x = 0$, then $z = 1$. The NOT operation is also referred to as the complement operation, since it changes a 1 to 0 and a 0 to 1, i.e., the result of complementing 1 is 0, and vice versa.

Binary logic (cont'd…)

- Binary logic resembles binary arithmetic, and the operations AND and OR have similarities to multiplication and addition, respectively.
- In fact, the symbols used for AND and OR are the same as those used for multiplication and addition.

Truth Tables of Logical Operations

AND

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

OR

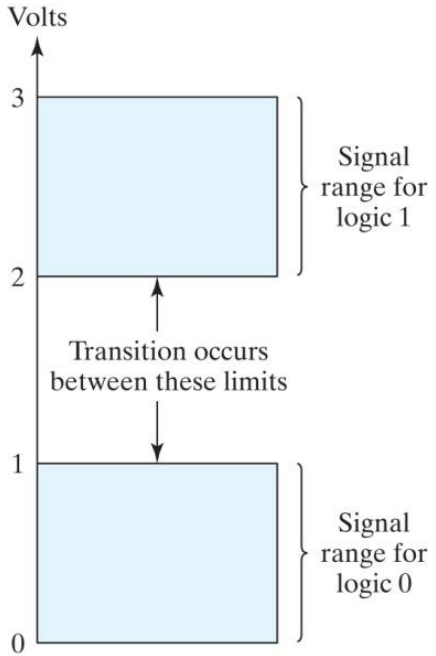
x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

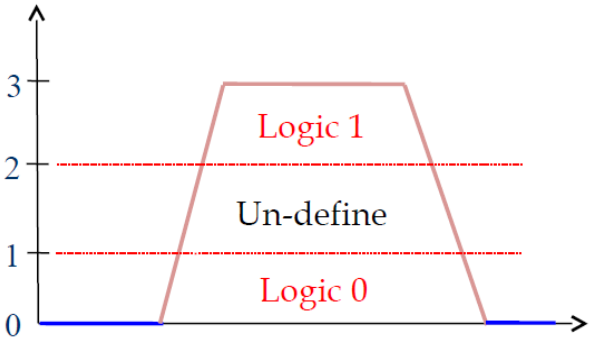
NOT

x	x'
0	1
1	0

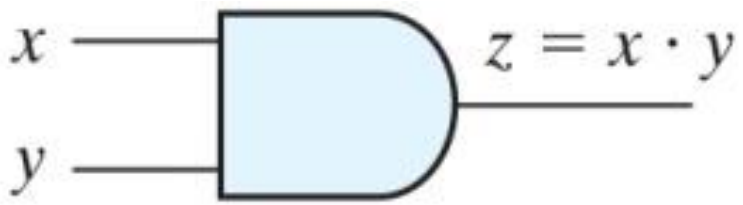
Binary logic (cont'd…)

- A "truth table" is a table of all possible combinations of the variables, showing the relation between the values that the variable may take and the result of the operation.
- Logic gates:
  - Logic gates are electronic circuits that operate on one or more input signals to produce an output signal.

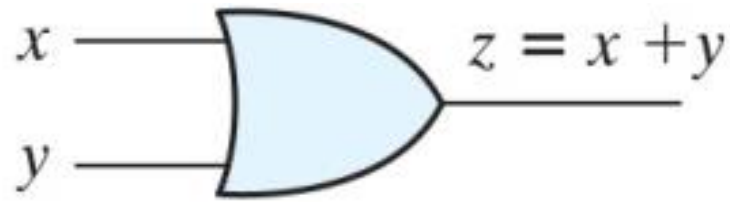




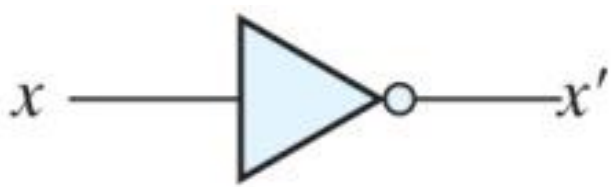
Logic gates (contd.)



Two-input AND gate



Two-input OR gate



NOT gate or inverter

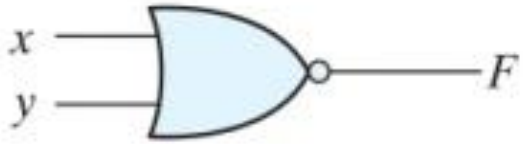
Logic gates (Contd.)

NAND

$F = (xy)'$

x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

NOR



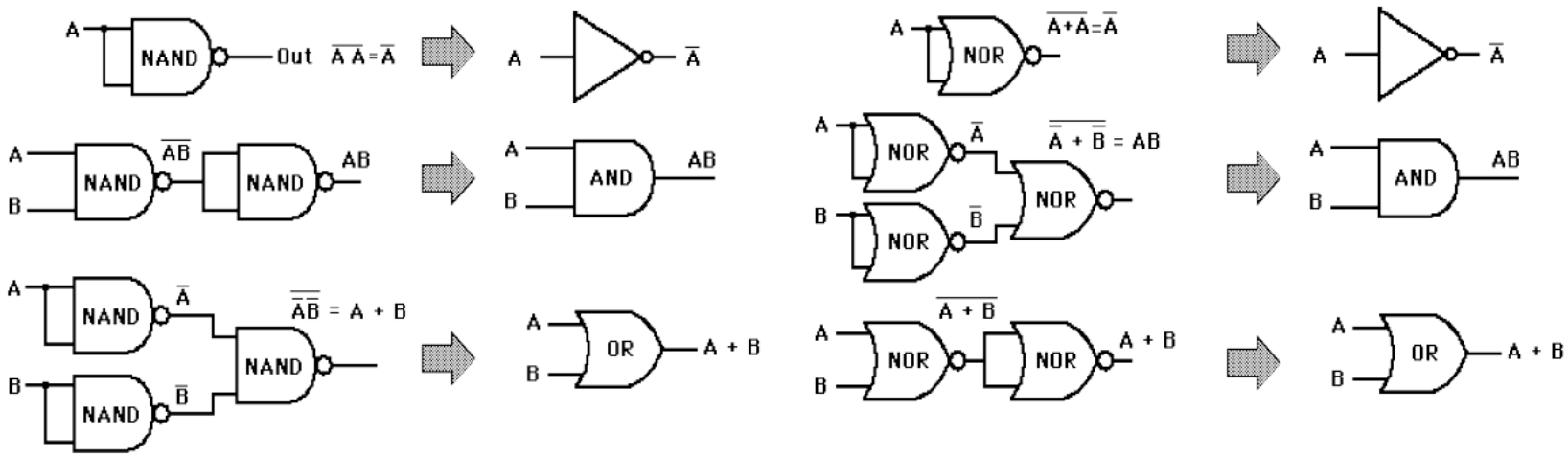
$F = (x + y)'$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

Universal Gate

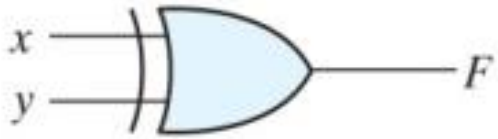
- **NAND and NOR** Gates are called ***Universal Gates*** because AND, OR and NOT gates can be implemented & created by using these gates.

NAND Gate Implementations NOR Gate Implementations



Logic gates (Cont'd.)

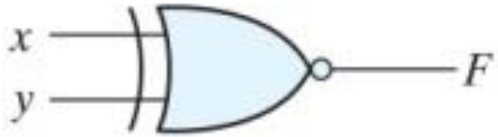
Exclusive-OR (XOR)



$$F = xy' + x'y$$
$$= x \oplus y$$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive-NOR or equivalence

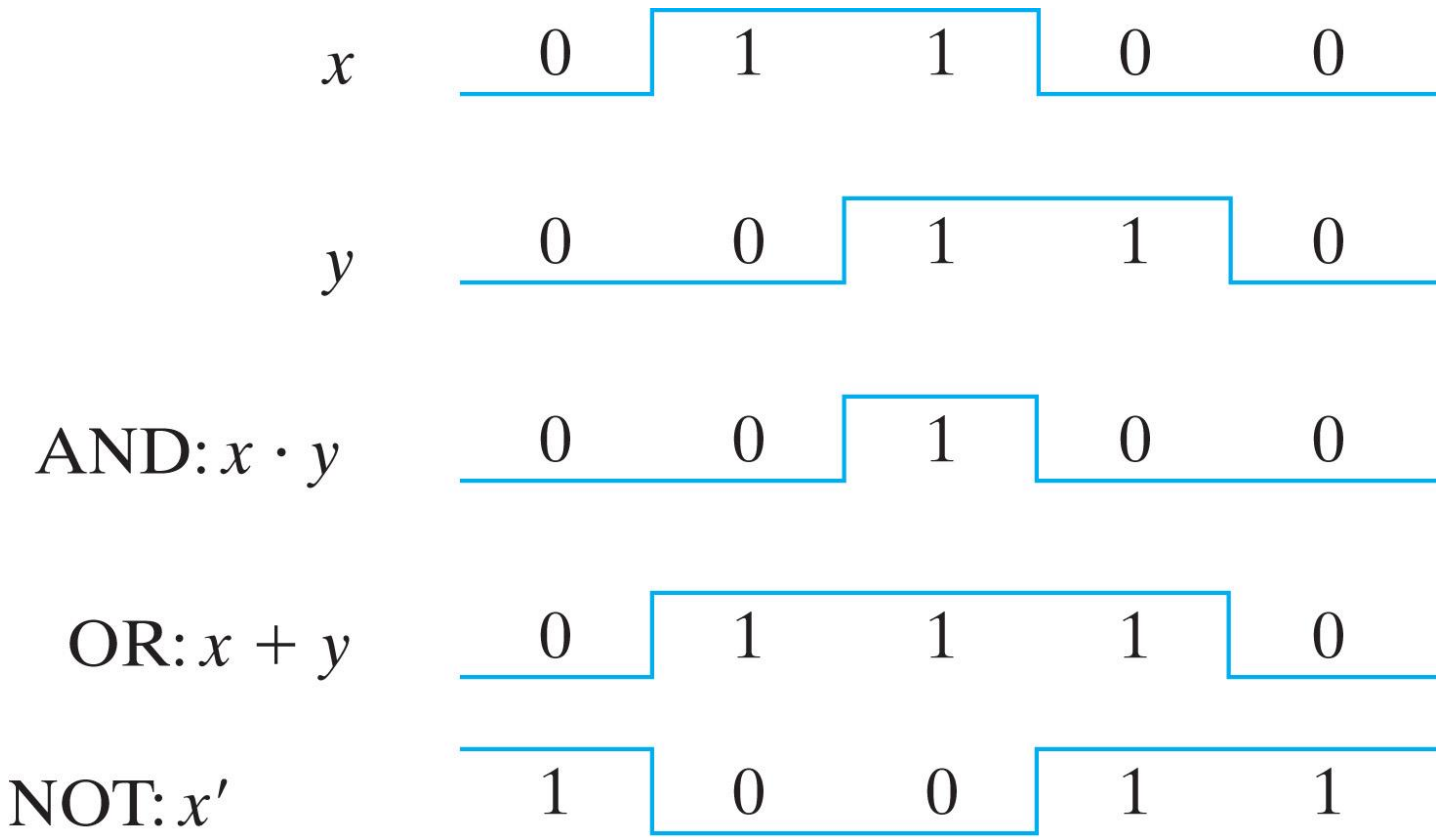


$$F = xy + x'y'$$
$$= (x \oplus y)'$$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1

Timing diagrams:

- Timing diagram illustrate the idealized response of each gate to the input signal combinations.



Binary Arithmetic

Binary Arithmetic

Addition

Decimal Addition

```
 1   1        ← Carry
     5   5
+    5   5
-----------
 1   1   0
```

= *Ten* ≥ *Base*

➔ Subtract a Base

Binary Addition

Column Addition

```
 1   1   1   1   1   1
     1   1   1   1   0   1     = 61
+        1   0   1   1   1     = 23
-------------------------------
 1   0   1   0   1   0   0     = 84
```

$\geq (2)_{10}$

Binary Arithmetic

Binary Subtraction

- Borrow a “base” when needed

		1			2			$= (10)_2$
	0	~~2~~	2	0	0	2		
	~~1~~	0	0	~~1~~	~~1~~	0	1	$= 77$
−			1	0	1	1	1	$= 23$
	0	1	1	0	1	1	0	$= 54$

Binary Arithmetic

Binary Multiplication

$$
\begin{array}{r}
1\ 0\ 1\ 1\ 1 \\
\times\quad 1\ 0\ 1\ 0 \\
\hline
0\ 0\ 0\ 0\ 0 \\
1\ 0\ 1\ 1\ 1 \\
0\ 0\ 0\ 0\ 0 \\
1\ 0\ 1\ 1\ 1 \\
\hline
1\ 1\ 1\ 0\ 0\ 1\ 1\ 0
\end{array}
$$

Binary Arithmetic

Binary Division

```
        110
11 ) 10010
      11
      ---
       11
       11
      ---
       00
        0
      ---
        0
```

Miscellaneous

Powers of 2

n	2^n	n	2^n	
0	$2^0=1$	8	$2^8=256$	
1	$2^1=2$	9	$2^9=512$	
2	$2^2=4$	10	$2^{10}=1024$	Kilo
3	$2^3=8$	11	$2^{11}=2048$	
4	$2^4=16$	12	$2^{12}=4096$	
5	$2^5=32$	20	$2^{20}=1M$	Mega
6	$2^6=64$	30	$2^{30}=1G$	Giga
7	$2^7=128$	40	$2^{40}=1T$	Tera

Thank you!!!